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Transbronchial Lung Cryobiopsy in Patients with Interstitial Lung Disease

A Systematic Review

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Abstract

Rationale: In 2018, a systematic review evaluating transbronchial lung cryobiopsy (TBLC) in patients with interstitial lung disease (ILD) was performed to inform American Thoracic Society, European Respiratory Society, Japanese Respiratory Society, and Asociación Latinoamericana del Tórax clinical practice guidelines on the diagnosis of idiopathic pulmonary fibrosis.

Objectives: To perform a new systematic review to inform updated guidelines.

Methods: Medline, Excerpta Medica Database, and the Cochrane Central Register of Controlled Trials (CCTR) were searched through June 2020. Studies that enrolled patients with ILD and reported the diagnostic yield or complication rates of TBLC were selected for inclusion. Data was extracted and then pooled across studies via meta-analysis. The quality of the evidence was appraised using the grading of recommendations, assessment, development, and evaluation approach.

Results: Histopathologic diagnostic yield (number of procedures that yielded a histopathologic diagnosis divided by the total number of procedures performed) of TBLC was 80% (95% confidence interval [CI], 76–83%) in patients with ILD. TBLC was complicated by bleeding and pneumothorax in 30% (95% CI, 20–41%) and 8% (95% CI, 6–11%) of patients, respectively. Procedure-related mortality, severe bleeding, prolonged air leak, acute exacerbation, respiratory failure, and respiratory infection were rare. The quality of the evidence was very low owing to the uncontrolled study designs, lack of consecutive enrollment, and inconsistent results.

Conclusions: Very low-quality evidence indicated that TBLC has a diagnostic yield of approximately 80% in patients with ILD, with manageable complications.

Keywords: cryobiopsy; interstitial lung disease; idiopathic pulmonary fibrosis; usual interstitial pneumonia

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Interstitial lung disease (ILD) collectively describes a group of diseases that cause inflammation and scarring in the lung parenchyma. When a patient presents with ILD, lung sampling is often indicated to help determine the specific cause of the disease in hope of initiating treatment that can arrest or slow its progression. Modalities available for lung tissue sampling include transbronchial forceps biopsy, transbronchial lung cryobiopsy (TBLC), and surgical lung biopsy.

Idiopathic pulmonary fibrosis (IPF) is the prototypical fibrosing ILD. The typical patient is a male, older than 60 years of age, usually with a previous history of smoking tobacco, who presents with insidious onset of cough and/or exertional dyspnea, bibasilar inspiratory crackles, and radiologic evidence of fibrosis predominantly in the lower lobes without an apparent cause.

In 2018, the American Thoracic Society (ATS), European Respiratory Society (ERS), Japanese Respiratory Society (JRS), and Asociación Latinoamericana del Tórax (ALAT) released clinical practice guidelines on the diagnosis of IPF (1), including the use of TBLC in patients with ILD who are clinically suspected of having IPF. In those guidelines, the committee recommended against TBLC in such patients if they have a high-resolution computed tomographic (HRCT) pattern of usual interstitial pneumonia (UIP) but did not make a definitive recommendation for or against TBLC in patients who have HRCT patterns other than UIP (i.e., probable UIP, indeterminate for UIP, etc.). The role of TBLC was recently reconsidered when the guidelines were updated, and this systematic review was performed to inform the guideline committee's recommendations regarding TBLC.

Methods

This review was performed in accordance with the guidance provided by the Cochrane Handbook for Systemic Reviews of Intervention (2). It was registered with the International Prospective Register of Systematic Reviews database (CRD42020208982).

Research Question

The research question was formulated using the population, intervention, comparator, outcome format: Should patients with newly detected ILD of unknown cause who are clinically suspected of having IPF undergo transbronchial lung cryobiopsy to obtain samples to make a histopathological diagnosis?

The critical outcome was diagnostic yield, while important outcomes included diagnostic agreement and various complications (procedural mortality, exacerbations, pneumothorax, bleeding, respiratory infection, and prolonged air leak). Regarding the selection of diagnostic yield rather than sensitivity and specificity as critical outcomes, diagnostic yield is appropriate if the intervention is the reference standard, but sensitivity and specificity are appropriate if the intervention is not the reference standard. In this case, histopathological diagnosis was chosen as the reference standard, making diagnostic yield the appropriate outcome. Clinical, radiological, and histopathological criteria applied by multidisciplinary discussion (MDD) were not chosen as the reference standard because this might lead to overestimates of the sensitivity and specificity due to "incorporation bias," which would have been misleading. Incorporation bias occurs when the test results are a component of the reference standard; in this case, histopathology obtained by TBLC is a component of the diagnostic criteria considered during MDD.

Literature Search

The guideline committee's medical librarian (S.L.K.) developed a sensitive search strategy and then searched Medline, Excerpta Medica Database, and Cochrane Central Register of Controlled Trials on the Ovid platform for relevant studies published between January 2016 and September 2020 (Table E1). January 2016 was selected as the starting date for the search because this was the final date of the search conducted for the previous ATS/ERS/JRS/ALAT guideline on the diagnosis of IPF (1). The search results were collected in a bibliographic database and then distributed to the methodology team.

Study Selection

A priori study selection criteria included *1*) enrolled patients with ILD of uncertain type in whom IPF was clinically suspected, *2*) evaluated TBLC, and *3*) reported diagnostic yield and/or procedural complications. Two methodologists (F.K. and J.P.U.B.) used a stepwise approach to screen the search results; they initially screened the titles and abstracts, then the full texts. Randomized trials that enrolled patients with ILD of uncertain type in whom IPF was clinically suspected and compared performing TBLC to not performing TBLC were sought first. Since no randomized trials were identified, observational studies (i.e., prospective cohort, retrospective cohort, case control, and before-and-after studies) were sought. Since no observational studies were identified, case series that enrolled at least 10 patients were sought. No studies were identified that specifically enrolled patients with ILD of uncertain type in whom IPF was clinically suspected; therefore, the selection criterion related to population was broadened to include indirect evidence. Specifically, studies were included if they enrolled patients with ILD of uncertain type (i.e., they did not need to specify that IPF was suspected). The stepwise approach of looking for randomized trials first, followed by controlled observational studies and then case series, was repeated. Case series with fewer than 10 patients, case reports, animal studies, and abstracts were excluded. When overlapping cohorts were identified, the older and invariably smaller cohort was excluded. Disagreements were resolved through discussion and consensus.

In addition to the stepwise approach described above, bibliographies of selected studies, systematic reviews, and review articles were reviewed for relevant studies. Guideline committee members suggested several studies that were reviewed for inclusion. Finally, studies that were included in the previous ATS/ERS/JRS/ALAT guideline on the diagnosis of IPF but predated and therefore were not identified by the current literature search were also selected (1).

Data Extraction

Data from the selected studies were extracted into a Microsoft Excel spreadsheet developed specifically for this review. The extracted information included the study setting, design and location, number of participants and their characteristics, type of ILD, intervention details, and outcomes. Information necessary to judge quality of evidence was also extracted from each study (3).

Evidence Synthesis

Data amenable to weighted pooling (i.e., meta-analysis) were analyzed using a random effects model in the Cochrane Collaboration Review Manager version 5.4.1

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software. Proportion was estimated using generic inverse variance. Individual values of 0 were replaced with 0.0001, and values of 1.0 were replaced with 0.9999. The 95% confidence interval (CI) was calculated for all summary estimates.

Statistical heterogeneity was measured using the I^2 statistic; I^2 values of 75% or higher were considered severe, 50–75% moderate, and 25–50% mild. The same approach was followed each time heterogeneity was encountered (4). First, the extracted data were confirmed. Second, whether to perform a meta-analysis was reconsidered. Third, sensitivity analyses were performed. The sensitivity analyses consisted of removing studies from the meta-analysis and, if the I^2 statistic improved, reviewing the full text of the removed studies to determine if those studies were similar to each other and different from the others in a way that might explain the different results. If a potential cause was found, subgroup analyses were performed. Finally, if no cause was found, outliers were eliminated. Outliers were defined as studies whose 95% CI did not overlap with the 95% CI of the summary estimate. Estimates after the elimination of outliers are reported in the tables and figures, whereas estimates before and after the elimination of outliers were presented to the guideline committee and are reported in the text.

Quality of Evidence Appraisal and Profile

The grading of recommendations assessment, development, and evaluation approach was used to appraise the quality of evidence and create the evidence profile. Briefly, a baseline assumption about the quality of the evidence was based on study design, then downgraded if any of the following were present: risk of bias (i.e., poor internal validity), inconsistency (i.e., heterogeneity of estimates across studies), indirectness (i.e., poor external validity), imprecision (i.e., wide 95% CI, few patients, or few events), and likelihood of publication bias (3). Regarding risk of bias, this was determined using a modified version of the Newcastle-Ottawa and Qadas-2 instruments. These instruments are intended for controlled observational studies but were modified to include only items relevant to studies without a control group, such as whether there was true diagnostic uncertainty among the participants (selection bias), consecutive enrollment (selection bias), and reporting of all outcomes (publication bias).

Results

A total of 202 articles were identified that addressed TBLC in patients with ILD. Most were excluded during the screening of titles and abstracts because they were not research studies, they enrolled a different population than desired, or they did not measure an outcome of interest. Fifty-six articles were retrieved for full text review, from which 34 articles were selected (5–38). An additional 5 articles that were included in the previous ATS/ERS/JRS/ALAT guideline on the diagnosis of IPF, but pre-dated the current literature search, were also selected (39–43), bringing the total to 39 studies that were selected for analysis (Figure 1). None of the studies compared the sampling procedure to not sampling. Rather, all studies reported various outcomes associated with sampling without including a control group (i.e., case series).

The studies ranged in size from 12 to 359 patients and used either a 1.9 or 2.4 mm cryoprobe with fluoroscopic guidance (5–43). Five of the studies were prospective (6, 18, 29, 30, 35). Studies reported using either general anesthesia or deep sedation. The number and locations of samples collected varied widely across studies. (Table 1).

Diagnostic Yield

Diagnostic yield was defined as the number of procedures that yielded a histopathologic diagnosis divided by the total number of procedures performed. It is worth emphasizing that it refers to any histopathologic diagnosis, whether an ILD or otherwise. Thirty-nine studies reported diagnostic yield (5–43), although one study was excluded from meta-analysis because it reported multiple values that could not be combined for inclusion in an aggregate data meta-analysis (38).

The diagnostic yield of TBLC in patients with ILD was 80% (95% CI, 76–84%), although there was serious inconsistency across studies. To look for sources of inconsistency, multiple subgroup analyses were performed, including publication date (before and after 2018 [Figure E2]), study size (more or less than 100 patients [Figure E3]), cryoprobe size (1.9 mm, 2.4 mm, or either [Figure E4]), and sample number (three or more samples, three samples, three or fewer samples, or unspecified [Figure E5]). Only sample number appeared to affect diagnostic yield, with a diagnostic yield of 85% (95% CI, 80–90%) when three or more samples were collected and a diagnostic yield of 77% or less when fewer samples were collected (Figure E5). Since the subgroup analyses were inadequate to explain the inconsistency, outliers were removed, bringing the I^2 statistic from 88% to 67%. Diagnostic yield was reassessed after the removal of outlying studies but remained nearly identical at 80% (95% CI, 76–83%) (Figure 2 and Table 2).

Agreement and Diagnostic Confidence

Two studies took patients with ILD of uncertain type, performed sequential TBLC and surgical lung biopsy (SLB) from the same lobe of the lung, and then assessed agreement between the diagnostic interpretation of TBLC samples and SLB samples (29, 35).

The larger ($n = 65$) study (i.e., the COLDICE study [Cryobiopsy versus Open Lung Biopsy in the Diagnosis of Interstitial Lung Disease Alliance]) demonstrated 70.8% agreement in histopathological diagnoses based on TBLC and SLB specimens, which increased to 76.9% diagnostic agreement following MDD. MDD diagnoses were made with high confidence in 60% using TBLC data, compared with 74% using SLB data ($P = 0.09$), with excellent concordance between the modalities (95%). *Post hoc* analysis of the study suggested that agreement can be improved by taking more samples, and that agreement would be higher if the diagnostic criteria for usual interstitial pneumonia are changed to favor central criteria (e.g., patchy fibrosis, fibroblastic foci, and absence of alternative features) over peripheral criteria (e.g., subpleural or paraseptal fibrosis) on lung biopsy (44).

In contrast, the smaller ($n = 21$) study (i.e., the Cryo-PID study) demonstrated agreement in the histopathological diagnoses derived from TBLC and SLB in only 8 out of 21 (38%) cases. The majority of diagnoses were usual interstitial pneumonia. Among 13 samples for which there was no agreement, TBLC was nondiagnostic and SLB was diagnostic in 4 cases, and TBLC and SLB yielded different diagnoses in 9 cases. Some of the different diagnoses were explainable by overlapping histopathological features. Diagnostic confidence was not measured (30).

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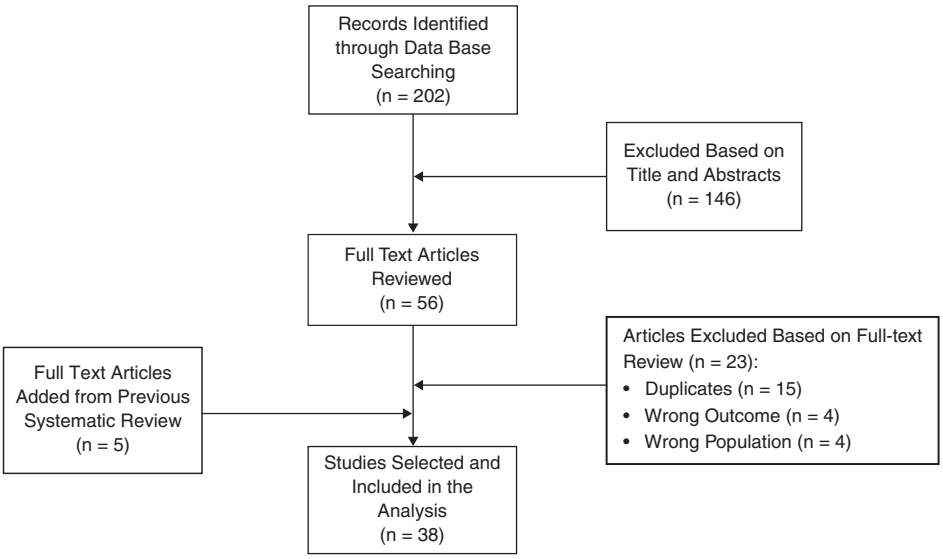

Figure 1. Flow of information diagram.

Pneumothorax

Thirty-three studies measured the pneumothorax rate as a complication (5–19, 21–25, 27–34, 37, 39–41, 43). Most performed routine chest radiographs 1 to 4 hours after the procedure. Pneumothoraces complicated 7% (95% CI, 6–8%) of procedures, although this estimate was inconsistent across studies. The cause of variability could not be identified; therefore, outlying studies were eliminated and the meta-analysis repeated (bringing the I^2 statistic from 95% to 60%). The estimate remained similar, with pneumothoraces complicating 8% (95% CI, 6–11%) of the procedures (Figure 3 and Table 2).

Bleeding

There was variation in the definition of bleeding across studies, but most used a qualitative scale. Twenty-eight TBLC studies reported any bleeding (5–13, 15, 16, 18, 19, 21, 22, 24, 25, 28, 29, 31–33, 35, 37, 39, 40, 42, 43), mild through severe, in 39% (95% CI, 23–55%) of patients, but the analysis was limited by inconsistent estimates across trials. The inconsistency could not be definitively explained but improved with the elimination of outlying studies (I^2 statistic decreased from 100 to 91%). After removal of the outlying studies, the estimated bleeding rate decreased to 30% (95% CI, 20–41%) (Figure 4 and Table 2). When the meta-analysis was repeated using only studies that reported using a bronchial blocker, the bleeding rate was 24% (95% CI, 14–34%)

Thirty studies reported severe bleeding (5, 7–16, 18, 20, 21, 23–25, 27, 28, 32–35, 37, 39–43, 45). Severe bleeding occurred in 40 patients. Given the rarity of events, the aggregated frequency of severe bleeding was negligible (0% [95% CI, 0–0%]) in patients undergoing TBLC, which was unchanged after one outlying study was removed (Table 2).

Mortality

Twenty studies reported mortality as a complication of TBLC; most studies did not specify a timeframe (5, 6, 9–12, 14–17, 21, 24, 28, 32–35, 37, 39, 43). Only three deaths were reported; two were attributed to acute exacerbations and one was unrelated to the procedure (28).

Exacerbation/Respiratory Failure

Fifteen studies assessed for acute exacerbations or respiratory failure that occurred after TBLC, reporting 31 events (6, 9, 11, 15, 19, 23, 24, 28, 29, 32–35, 37, 43). Given the rarity of the events, the aggregated frequency of acute exacerbations or respiratory events was negligible (0% [95% CI, 0–0%]) in patients undergoing TBLC and remained the same after one outlying study was removed (Figure E5 and Table 2).

Respiratory Infection

Nine studies assessed for respiratory infections occurring after TBLC, reporting 9 events (6, 9, 11, 19, 22, 24, 28, 29, 39). Given the rarity of the events, the aggregated frequency of respiratory infections was negligible (0% [95% CI, 0–0%]) in patients undergoing TBLC, which remained the same after one outlying study was removed (Figure E6 and Table 2). Types of respiratory infection included lower respiratory tract infections.

Persistent Air Leak

Six studies reported the frequency of persistent air leak after TBLC, reporting four events (8, 9, 11, 14, 29, 39). Given the rarity of the events, the aggregated frequency of persistent air leak was negligible (0% [95% CI, 0–0%]) in patients undergoing TBLC (Figure E7 and Table 2).

Quality of Evidence

The quality of evidence (i.e., confidence in the estimated effects) was very low for all outcomes. Nonrandomized studies without a control group (i.e., case series) begin with an assumption of very low-quality evidence. Most individual studies had serious risk of bias due to lack of consecutive enrollment, and many outcomes were inconsistent across studies.

Discussion

This systematic review was conducted to inform an update of the 2018 ATS/ERS/JRS/ALAT clinical practice guidelines on the diagnosis of IPF (1). The systematic review indicates that TBLC has a diagnostic yield of 80% (95% CI, 76–83%) in patients with ILD. It appears to be a relatively safe procedure,

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Table 1. Characteristics of included studies

Author	Year	Location	Duration	Patients	Setting/ Design	Cryoprobe	Samples	Type of Sedation	Fluoroscopy	Risk of Bias
Abdelghani	2019	United States	N/A	40	Retrospective	1.9 mm	Unknown	General anesthesia	Yes	Severe
Aburto	2020	Spain	5 yr	257	Prospective	1.9 mm/2.4 mm	Unknown	General anesthesia	Yes	Severe
Aragaki-Nakahodo	2017	United States	3 yr	36	Retrospective	1.9 mm/2.4 mm	5	General anesthesia	Yes	Very severe
Bango-Álvarez	2017	Spain	2 yr	106	Retrospective	1.9 mm	3	Deep sedation	Yes	None
Bondue	2017	Belgium	1 yr	30	Retrospective	1.9 mm/2.4 mm	2–5	General anesthesia	Yes	None
Camuset	2019	France	3 yr	24	Retrospective	1.9 mm/2.4 mm	3	General anesthesia	Yes	Severe
Cascante	2016	Spain	3 yr	55	Retrospective	2.4 mm	2–3	General anesthesia	Yes	None
Cho	2019	United States	3 yr	40	Retrospective	1.9 mm	3–5	Deep sedation	Yes	Severe
Cirak	2020	Turkey	N/A	82	Retrospective	1.9 mm/2.4 mm	Unknown	General anesthesia	Yes	Very severe
Cooley	2018	United States	4 yr	159	Retrospective	1.9 mm/2.4 mm	Unknown	General anesthesia	Yes	Severe
Dhooria	2018	India	2 yr	128	Retrospective	1.9 mm	3	Deep sedation/general anesthesia	Yes	None
Echevarria-Uraga	2016	Spain	2 yr	100	Retrospective	2.4 mm	Unknown	General anesthesia	Yes	Very severe
Fruchter	2014	Israel	2 yr	75	Retrospective	1.9 mm	1–2	Deep sedation	Yes	Severe
Griff	2014	Germany	2 yr	52	Retrospective	1.9 mm	3–4	Deep sedation	Yes	Very severe
Hagmeyer	2016	Germany	1 yr	51	Retrospective	1.9 mm/2.4 mm	3–5	Deep sedation	Yes	Very severe
Hernandez-Gonzalez	2015	Spain	3 yr	33	Retrospective	1.9 mm	3	General anesthesia	Yes	Severe
Hetzel	2019	Germany	N/A	359	Prospective	1.9 mm/2.4 mm	Unknown	Deep sedation/general anesthesia	Yes	Very severe
Inomata	2020	Japan	1 yr	112	Retrospective	1.9 mm/2.4 mm	2–3	General anesthesia	Yes	Severe
Jacob	2019	Portugal	3 yr	32	Retrospective	1.9 mm/2.4 mm	3	General anesthesia	Yes	Very severe
Kronborg-White	2017	Denmark	1 yr	38	Retrospective	1.9 mm/2.4 mm	4	General anesthesia	Yes	Severe
Kropski	2013	United States	1 yr	25	Retrospective	1.9 mm	1–2	Deep sedation	Yes	Very severe
Lentz	2018	United States	3 yr	104	Retrospective	1.9 mm	2–3	Deep sedation	Yes	Very severe
Linhas	2017	Portugal	2 yr	90	Retrospective	2.4 mm	Unknown	General anesthesia	Yes	None
Pajares	2014	Spain	N/A	39	Retrospective	2.4 mm	3–4	General anesthesia	Yes	None
Pourabdollah	2014	Iran	N/A	40	Retrospective	2.4 mm	2–3	Deep sedation	Yes	Severe
Pourabdollah	2016	Iran	N/A	41	Retrospective	2.4 mm	1	Deep sedation	Yes	Very severe
Ramaswamy	2016	United States	1 yr	56	Retrospective	2.4 mm	1	Deep sedation	Yes	Severe
Ravaglia	2019	Italy	6 yr	699	Retrospective	1.9 mm/2.4 mm	1–11	Deep sedation	Yes	Severe
Ravaglia	2017	Italy	4 mo	45	Prospective	2.4 mm	4	Deep sedation	Yes	None
Romagnoli	2019	Italy	2 yr	21	Prospective	2.4 mm	2–6	General anesthesia	Yes	Severe
Shafiek	2019	Egypt	1 yr	12	Retrospective	2.4 mm	2–4	Deep sedation	Yes	None
She	2020	Australia	N/A	121	Retrospective	1.9 mm/2.4 mm	1–7	Deep sedation/general anesthesia	Yes	None
Sriprasart	2017	Thailand	2 yr	74	Retrospective	1.9 mm/2.4 mm	5	Deep sedation	Yes	Severe
Tomassetti	2016	Italy	1 yr	58	Retrospective	2.4 mm	1	Deep sedation	Yes	None
Troy	2020	Australia	N/A	65	Prospective	1.9 mm/2.4 mm	3–7	General anesthesia	Yes	None
Unterman	2019	Israel	8 yr	14	Retrospective	2.4 mm	3–7	Deep sedation	Yes	Very severe
Ussavarungsi	2017	United States	2 yr	74	Retrospective	1.9 mm	3	Deep sedation	Yes	Very severe
Wälscher	2019	Germany	2 yr	109	Retrospective	1.9 mm/2.4 mm	1–8	General anesthesia	Yes	None
Tomassetti	2020	Italy	3 yr	266	Retrospective	2.4 mm	1	Deep sedation	Yes	None

Definition of abbreviation: N/A = not applicable.

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Study or Subgroup	Diagnostic yield	SE	Weight	Diagnostic yield IV, Random, 95% CI
Abdelghani 2019	0.93	0.04034	4.6%	0.93 [0.85, 1.01]
Aburto 2020	0.8	0.02495	5.6%	0.80 [0.75, 0.85]
Aragaki-Nakahodo 2017	0.81	0.06538	3.1%	0.81 [0.68, 0.94]
Bango-Álvarez 2017	0.86	0.0337	5.1%	0.86 [0.79, 0.93]
Bondue 2017	0.8	0.07303	2.8%	0.80 [0.66, 0.94]
Camuset 2019	0.64	0.09798	1.9%	0.64 [0.45, 0.83]
Cascante 2016	0.87	0.04535	4.3%	0.87 [0.78, 0.96]
Cho 2019	0.85	0.05646	3.6%	0.85 [0.74, 0.96]
Cooley 2018	0.69	0.03668	4.9%	0.69 [0.62, 0.76]
Dhooria 2018	0.78	0.03661	4.9%	0.78 [0.71, 0.85]
Griff 2014	0.77	0.05836	3.5%	0.77 [0.66, 0.88]
Hagmeyer 2016	0.75	0.06063	3.4%	0.75 [0.63, 0.87]
Hernandez-Gonzalez, 2015	0.79	0.0709	2.9%	0.79 [0.65, 0.93]
Jacob 2019	0.78	0.07323	2.8%	0.78 [0.64, 0.92]
Kronborg-White 2017	0.84	0.05947	3.4%	0.84 [0.72, 0.96]
Kropski 2013	0.8	0.08	2.5%	0.80 [0.64, 0.96]
Linhas 2017	0.78	0.04367	4.4%	0.78 [0.69, 0.87]
Pajares 2014	0.74	0.07024	2.9%	0.74 [0.60, 0.88]
Ramaswamy 2016	0.66	0.0633	3.2%	0.66 [0.54, 0.78]
Romagnoli 2019	0.81	0.08561	2.3%	0.81 [0.64, 0.98]
Shafiek 2019	0.83	0.10844	1.6%	0.83 [0.62, 1.04]
She 2020	0.66	0.04306	4.4%	0.66 [0.58, 0.74]
Sriprasart 2017	0.88	0.03778	4.8%	0.88 [0.81, 0.95]
Tomassetti 2016	0.91	0.03758	4.8%	0.91 [0.84, 0.98]
Troy 2020	0.89	0.03881	4.7%	0.89 [0.81, 0.97]
Unterman 2019	0.93	0.06819	3.0%	0.93 [0.80, 1.06]
Wälscher 2019	0.73	0.04252	4.5%	0.73 [0.65, 0.81]
Total (95% CI)			**100.0%**	**0.80 [0.77, 0.84]**

Heterogeneity: Tau^2 = 0.00; Chi^2 = 70.47, df = 26 (P < 0.00001); I^2 = 63%
Test for overall effect: Z = 49.99 (P < 0.00001)

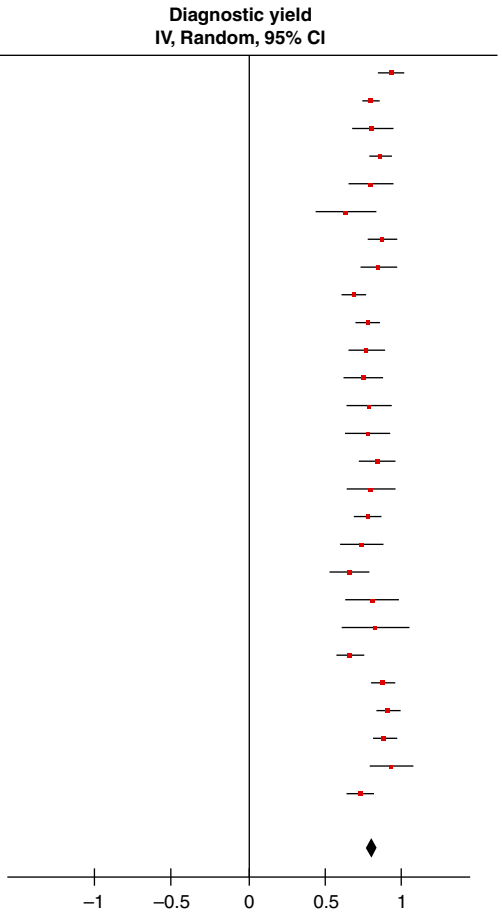

Figure 2. Diagnostic yield meta-analysis. CI = confidence interval; df = degree of freedom; I^2 = heterogeneity index; IV = inverse variance; SE = standard error.

causing mostly minor bleeding in 30% (95% CI, 20–41%) and pneumothorax in 8% (95% CI, 6–11%). Mortality, severe bleeding, prolonged air leaks, acute exacerbations, respiratory failure, and respiratory infections are rare.

These results invite the question, "Is transbronchial lung cryobiopsy an acceptable alternative to surgical lung biopsy for making a histopathological diagnosis in patients with ILD of undetermined type?" Key considerations in answering the question include *1*) the estimated diagnostic yield of TBLC and SLB are 80% and 90% (1), respectively; *2*) the sampling techniques provide similar diagnostic confidence in the context of an MDD; and *3*) TBLC is both less invasive and less costly than SLB.

Compared with the systematic review that informed the prior guideline, the current systematic review includes an additional 26 studies (1,426 patients). The estimated diagnostic yield is identical, but the 95% CI has narrowed (from 80% [95% CI, 74–86%] to 80% [95% CI] 76–83%). The estimated rate of bleeding is higher in the current systematic review, probably reflecting the threshold used to define bleeding and due to a broader utilization of TBLC including nonspecialized centers, while all other complication estimates are lower, possibly reflecting greater experience among those who perform TBLC (1).

It is noteworthy that both this systematic review and the systematic review that informed the prior diagnosis of IPF guideline (1) are similar to a systematic review that was performed to inform ATS/JRS/ALAT diagnosis of hypersensitivity pneumonitis (HP) guidelines (46). Whereas the diagnosis of IPF systematic reviews were intended for patients with ILD of unknown type in whom IPF is clinically suspected, the systematic review performed for the HP guideline was intended for patients with ILD of unknown type that is consistent with possible HP. Despite the different populations of interest, neither the systematic reviews performed for the diagnosis of IPF guidelines nor the systematic review performed for the diagnosis of HP guidelines identified any studies that specifically enrolled patients with suspected IPF or possible HP, respectively. The guideline committees chose to consider indirect evidence, and therefore, studies that enrolled patients with ILD of unknown type were selected, without requiring that the patients have either suspected IPF or possible HP. The result was that the same body of evidence was synthesized.

Compared with systematic reviews completed by other investigators looking at TBLC in ILD, the current results are similar. In 2017, Sharp and colleagues reported a diagnostic yield of 84%, pneumothorax rate of 10%, moderate bleeding rate of 21%, and procedural mortality rate of 0.5% (47). In 2016, Johannson and colleagues reported a diagnostic yield of 83%, pneumothorax rate of 12%, severe bleeding rate of 39%, and procedural mortality rate of 0.003% (48).

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Table 2. Evidence profile for transbronchial cryobiopsy in patients with fibrotic interstitial lung disease

Quality Assessment							Summary of Findings			
Studies, *n*	Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other	Patients	Effect after Outliers Removed (95% CI)	Quality	Importance
Diagnostic yield										
29*	Case series	Very severe†	Very Severe‡	None	None	None	1846	0.80 (0.76–0.83)	⊕OOO VERY LOW	CRITICAL
Mortality, 30-d										
20§	Case series	Very severe†	None	None	None	None	2231	0.00 (0.00–0.00)	⊕OOO VERY LOW	IMPORTANT
Bleeding, all										
12‖	Case series	Very severe†	Very Severe‡	None	None	None	617	0.30 (0.20–0.41)	⊕OOO VERY LOW	IMPORTANT
Bleeding, severe										
29¶	Case series	Very severe†	None	None	None	None	2910	0.00 (0.00–0.00)	⊕OOO VERY LOW	IMPORTANT
Pneumothorax										
20**	Case series	None	Very Severe‡	None	None	None	1796	0.08 (0.06–0.11)	⊕OOO VERY LOW	IMPORTANT
Prolonged air leak										
6††	Case series	Very severe†	None	None	None	None	469	0.00 (0.00–0.00)	⊕OOO VERY LOW	IMPORTANT
Exacerbations										
14‡‡	Case series	Very severe†	Severe§§	None	None	None	1863	0.00 (0.00–0.00)	⊕OOO VERY LOW	IMPORTANT
Respiratory infections										
7‖‖	Case series	Very severe†	None	None	None	None	444	0.00 (0.00–0.00)	⊕OOO VERY LOW	IMPORTANT

Definition of abbreviation: CI = confidence interval.
*Studies: (5–10, 12, 14, 15, 17, 21, 22, 24–27, 30–37, 39, 41–43).
†Risk of bias owing to lack of consecutive enrollment and true diagnostic uncertainty.
‡Inconsistency indicated by I^2 of more than 75% before outliers' removal.
§Studies: (5, 6, 9–12, 14–17, 21, 24, 28, 32–35, 37, 39, 43).
‖Studies: (5, 7, 10, 11, 13, 21, 22, 25, 29, 35, 37, 42).
¶Studies: (5, 7–15, 18, 21, 23–25, 27–29, 32–35, 37, 39, 41–43).
**Studies: (5, 7–9, 13–16, 18, 19, 21, 23–25, 29, 30, 33, 37, 39, 40).
††Studies: (8, 9, 11, 14, 29, 39).
‡‡Studies: (6, 9, 11, 19, 23, 24, 28, 29, 32–35, 37, 43).
§§Inconsistency indicated by I^2 of 50–75% before outliers' removal.
‖‖Studies: (9, 11, 19, 22, 24, 29, 39).

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Study or Subgroup	PTX	SE	Weight	PTX IV, Random, 95% CI
Abdelghani 2019	0.05	0.03446	5.1%	0.05 [−0.02, 0.12]
Aragaki-Nakahodo 2017	0.11	0.05215	3.1%	0.11 [0.01, 0.21]
Bango-Álvarez 2017	0.05	0.02117	7.3%	0.05 (0.01, 0.09]
Bondue 2017	0.2	0.07303	1.9%	0.20 [0.06, 0.34]
Cascante 2016	0.15	0.04815	3.5%	0.15 (0.06, 0.24]
Cirak 2020	0.07	0.02818	6.1%	0.07 [0.01, 0.13]
Cooley 2018	0.11	0.02481	6.6%	0.11 [0.06, 0.16]
Dhooria 2018	0.11	0.02766	6.2%	0.11 [0.06, 0.16]
Echevarria-Uraga 2016	0.03	0.01706	8.0%	0.03 [−0.00, 0.06]
Fruchter 2014	0.03	0.0197	7.6%	0.03 [−0.01, 0.07]
Hetzel 2019	0.08	0.02059	7.4%	0.08 [0.04, 0.12]
Inomata 2020	0.05	0.06481	2.3%	0.05 [−0.08, 0.18]
Jacob 2019	0.16	0.06481	2.3%	0.16 [0.03, 0.29]
Lentz 2018	0.03	0.01673	8.1%	0.03 [−0.00, 0.06]
Linhas 2017	0.22	0.04367	3.9%	0.22 (0.13, 0.31]
Pajares 2014	0.08	0.04344	4.0%	0.08 [−0.01, 0.17]
Ravaglia 2017	0.16	0.05465	2.9%	0.16 [0.05, 0.27]
Romagnoli 2019	0.1	0.06547	2.3%	0.10 [−0.03, 0.23]
Sriprasart 2017	0.07	0.02966	5.8%	0.07 [0.01, 0.13]
Wälscher 2019	0.12	0.03113	5.6%	0.12 [0.06, 0.18]
Total (95% CI)			**100.0%**	**0.08 [0.06, 0.11]**

Heterogeneity: Tau^2 = 0.00; Chi^2 = 47.32, df = 19 (P = 0.00003); I^2 = 60%
Test for overall effect: Z = 7.55 (P < 0.00001)

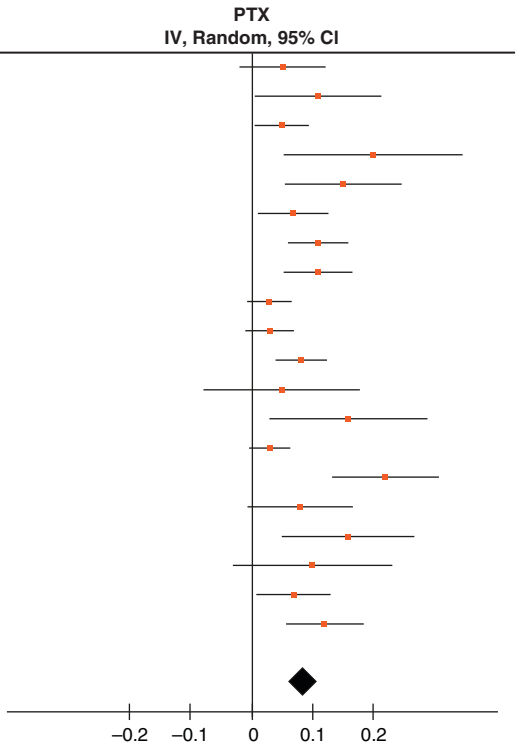

Figure 3. Pneumothorax meta-analysis. CI = confidence interval; df = degree of freedom; I^2 = heterogeneity index; IV = inverse variance; PTX = pneumothorax; SE = standard error.

Study or Subgroup	All Type of Bleeding	SE	Weight	All Type of Bleeding IV, Random, 95% CI
Abdelghani 2019	0.28	0.07099	8.1%	0.28 [0.14, 0.42]
Aragaki-Nakahodo 2017	0.67	0.07837	7.9%	0.67 [0.52, 0.82]
Camuset 2019	0.21	0.08314	7.7%	0.21 [0.05, 0.37]
Cirak 2020	0.48	0.05517	8.6%	0.48 [0.37, 0.59]
Hernandez-Gonzalez, 2015	0.31	0.08051	7.8%	0.31 [0.15, 0.47]
Jacob 2019	0.03	0.03016	9.2%	0.03 [−0.03, 0.09]
Kronborg-White 2017	0.16	0.05947	8.5%	0.16 [0.04, 0.28]
Pajares 2014	0.56	0.07949	7.8%	0.56 [0.40, 0.72]
Ravaglia 2017	0.3	0.06831	8.2%	0.30 [0.17, 0.43]
Troy 2020	0.22	0.05138	8.7%	0.22 [0.12, 0.32]
Ussavarungsi 2017	0.22	0.04816	8.8%	0.22 [0.13, 0.31]
Wälscher 2019	0.28	0.04301	8.9%	0.28 [0.20, 0.36]
Total (95% CI)			**100.0%**	**0.30 [0.20, 0.41]**

Heterogeneity: Tau^2 = 0.03; Chi^2 = 118.58, df = 11 (P < 0.00001); I^2 = 91%
Test for overall effect: Z = 5.70 (P < 0.00001)

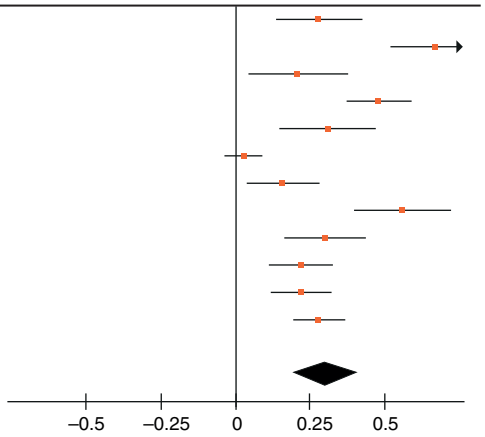

Figure 4. Bleeding (all, mild to severe) meta-analysis. CI = confidence interval; df = degree of freedom; I^2 = heterogeneity index; IV = inverse variance; SE = standard error.

Moreover, in 2019, Ravaglia and colleagues reported data from 699 patients from a high-volume, experienced center, which included a diagnostic yield of 87.8%, pneumothorax rate of 19.2%, no severe bleeding, and procedural mortality rate of 0.4% (28).While most studies indicate that the rate of complications is modest in the hands of experienced operators, it is important to be aware that the risk of complications, including life-threatening bleeding, might be higher when first introduced into clinical practice or performed with less experienced bronchoscopists (49).

Two studies assessed agreement between TBLC and surgical lung biopsy specimens from patients with ILD of uncertain type who underwent sequential TBLC and SLB from the same lobe of the lung, followed by MDD. Participants were blinded to the sampling technique and tried to eventually reach a consensus on clinical diagnosis. The COLDICE study provided the largest dataset of histopathological tissue obtained from both TBLC and SLB for ILD diagnosis. The histopathological agreement between TBLC and SLB was 70.8%, whereas diagnostic agreement at MDD was 76.9% (35). In contrast, the Cryo-PID study demonstrated only 38% agreement in the histopathological

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diagnoses (30), which is considerably poorer than the COLDICE study. Potential reasons for the discordant results have been attributed to methodological differences, including whether *1*) the MDD was centralized or in the participating centers, *2*) histopathological diagnoses were made by a single expert pathologist or the consensus of three expert pathologists, and *3*) the kappa or weighted kappa statistic was used to measure agreement (50, 51). Regarding the choice of statistic, the kappa (reported by the Cryo-PID study [30]) tends to be lower than the weighted kappa (reported by the COLDICE study [35]) because the former accounts for discordance alone, while the latter also accounts for degree of the discordance. Kappa is an appropriate index of agreement when the categories are nominal, and the weighted kappa is an appropriate index of agreement when the categories are ordinal or ranked.

Strengths and Limitations

The major strength of this systematic review is that it was part of guideline development. A multidisciplinary international committee of experts ensured that the question and outcomes were relevant to practicing clinicians and used their experience to interpret the clinical significance of the estimated effects. In addition, the experts were aware of procedural details that helped in the evaluation of potential causes of inconsistency across studies.

The primary limitation of the systematic review is the quality of the evidence. Confidence in the estimated effects is very low due to the uncontrolled study designs, risk of bias due to lack of consecutive enrollment in many studies, and frequent inconsistent estimates across studies. Randomized trials or controlled observational studies are needed to compare clinical outcomes among patients who undergo TBLC to patients who do not undergo TBLC and, instead, proceed directly to SLB. Another limitation pertains to diagnostic yield being an imperfect outcome. There are many aspects of TBLC that might affect the diagnostic yield, and, therefore, the outcome is highly susceptible to variation across studies. As examples, procedure volume, experience of the operator, number of specimens, and the threshold of the pathologist to commit to a diagnosis may all influence diagnostic yield. Until TBLC is fully standardized, the estimated diagnostic yield is susceptible to inconsistent estimates. ■

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